21-701: Discrete Mathematics

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For the latest version, visit https://thefundamentaltheor3m.github.io/DiscreteMathNotes/main.pdf.

For comments, errors or suggestions, please raise an issue/make a pull request at https://github.com/thefundamentaltheor3m/DiscreteMathNotes.

Contents

Course Introduction and Overview

This course will cover the theory of absolutely everything mathematical.

Chapter 1

Graphs and Colourings

This is where you write something generic about the thing you're introducing. In my case, that would be introductions. Except my work is cut out for me, because the concept of introductions needs no introduction...

Notation.

- $[k]$ will denote $\{1, \ldots, k\}$.
- $\sim$ will denote the adjacency relation on graph vertices.

1.1 Graph Colourings

Definition 1.1.1 (Colouring)**.** Given a graph $G = (V, E)$, a k-colouring of G is a map $c : V \to [k]$ such that $u \sim v \implies c(u) \neq c(v)$.

1.1.1 2-Colourings

Let's start with a motivating question: which graphs can be 2-coloured?

Any odd cycle graph can't be 2-coloured. That is, any graph that looks like

Let's study when odd cycles exist.

Lemma 1.1.2. *If G has an odd closed walk, then G has an odd cycle.*

Recall that a closed walk need not be a cycle: walks can contain repetitions. So even line graphs can have closed walks (start from one vertex, go to its neighbour, return).

Proof. Suppose G has an odd closed walk. Let W be such a walk of minimal length. Write $W = (x_1, \ldots, x_i, \ldots, x_j, \ldots, x_\ell)$, with $x_\ell = x_1$. Suppose W is not a cycle. Then there must be some x_i, x_j such that $x_i = x_j$ even though $i \neq j$, ie, there must be some repetition. Then, $(x_1, \ldots, x_i, x_{j+1}, \ldots, x_\ell)$ and $(x_i, x_{i+1}, \ldots, x_j)$ are both shorter walks than W. Moreover, one of them must be of odd length. This contradicts minimality of the length of W. So W must be a cycle. $\square$

Here's a cool result.

Theorem 1.1.3. *A graph $G = (V, E)$ can be 2-coloured if and only if G has no odd cycle.*

Proof.

($\implies$) We've already seen that odd cycles can't be 2-coloured. So if G can be 2-coloured, it can't have an odd cycle. We've essentially seen this from the examples above.

($\impliedby$) Fix some $v_0 \in V$. Define $c : V \to \{1, 2\}$ by

$$c(u) = \begin{cases} 2 & \text{if } \operatorname{dist}(v_0, u) \text{ is odd} \\ 1 & \text{if } \operatorname{dist}(v_0, u) \text{ is even} \end{cases}$$

where $\operatorname{dist}(\cdot, \cdot)$ denotes the shortest number of edges from one vertex to another.

We claim that this c is a valid 2-colouring.

Indeed, suppose it is not. Then, there must be some $u, v \in V$ such that $u \sim v$ and $c(u) = c(v)$. Now let's consider the two possible cases on the colour.

Suppose $c(u) = c(v) = 1$ for adjacent u, v. Then, there is a path of odd length between v and v_0 and another path of odd length between u and v_0. Then, concatenating the paths v

to v_0 to u, we get an odd cycle.

□

Definition 1.1.4 (Hypergraph). Given a vertex set V, a **hypergraph** is a collection H of subsets of V. The sets in H are called **hyperedges**.

Definition 1.1.5 (Uniformity). A hypergraph H is k-**uniform** if $|f| = k$ for all $f \in H$.

The following is thus obvious.

Proposition 1.1.6. *A graph is a 2-uniform hypergraph.*

Definition 1.1.7. A proper colouring of a hypergraph H is an assignment $c : V \to [r]$ such that $\forall f \in H$, $|c(f)| \geq 2$.

We say "no edge is monochromatic".

We can ask ourselves the following question: **"Which 3-uniform hypergraphs are 2-colourable?**

For each k, define $m(k)$ to be the smallest m such that $\exists$ some k-uniform hypergraph with m edges that is not 2-colourable. We can show that $m(2) = 3$. But what is $m(3)$? That is, how many 3-edges do I need to prevent 2-colourability?

Let H be a 3-uniform hypergraph that's not 2-colourable. Assume, further, that H has at most 6 edges (each being a subset of size 3).

First, we'll show that we can reduce, without loss of generality, to the case where H has exactly 6 vertices. Indeed, if H has vertices x, y that are not in any common edge, then I can define H' by "merging" x and y. So H' has one fewer vertex than H. This H' cannot be 2-colourable: if it were, then that colouring would extend to H by using the colour of the merged vertex at both x and y, which would make H 2-colourable. One can show that when H has more than 6 vertices, it is always possible to find such a pair of vertices sharing no common edge.

So now assume that H has exactly 6 edges and 6 vertices and is not 2-colourable.

1.2 Another Section

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Chapter 2

Another Chapter

You get the idea.

2.1 Introducing the Main Object of Study in this Chapter

Woah. Very cool.

2.2 Another Section

Yup, `\lipsum` time. Boy do I love LaTeX!

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